Supreme
4047192241

2026-06-21 19:00 (UTC+8)

Contingency Contract Sharing Code: 001008986W2A0JJ

Sample Space :

Toss a coin $\rightarrow$ $\Omega = \{\overset{\omega_1}{HH}, \overset{\omega_2}{HT}, \overset{\omega_3}{TH}, \overset{\omega_4}{TT}\}$

$P \rightarrow p_1, p_2\ p_3\ p_3$

$\Sigma p_i = 1$

$p_i \geq 0$

event $A = \{$At least 1 T appears$\}$

$= \{HT, TH, TT\}$

$P(A) = p_2 + p_3 + p_4 = 1 - p_1$

$= 1 - P(A')$

$\mathcal{F} = \{$subsets of $\Omega\}$ $\rightarrow$ <u>σ-field</u>

A random variable X is a function $f : \Omega \rightarrow (-\infty, \infty)$

$X = \#$ of heads

$X(\omega_1) = 2$

$X(\omega_2) = 1$

$X(\omega_3) = 1$

$X(\omega_4) = 0$

$X_2 = 5X$

$X_3 = \# H's - \# T's$

$X \sim Bin(n,p)$, X = H's in n tosses where $p = P(H)$

$$P(X=k) = {}^nC_k \, p^k (1-p)^{n-k} \text{ for } k = 0, \dots, n$$

$X \sim$ Bernoulli $(p) \leftrightarrow X \sim$ Binomial $(1,p)$

Y = longest run of H's

$X_1, X_2, \dots$ on Ω $= \{X_i : i \in \mathbb{N}\}$ → discrete

$\{X_t : t \geq 0\}$ → continuous

Stochastic Process → Sequence of these random variables

$F(x) = P(X \leq x)$ → Cumulative Distribution Function CDF

X_1, X_2 are independent if $P(X_1 \leq x_1, X_2 \leq x_2) = P(X_1 \leq x_1) . P(X_2 \leq x_2)$

$X_1, X_2, \dots X_n$ are mutually independent if

$$P(X_1 \leq x_1, X_2 \leq x_2, \dots, X_n \leq x_n) = P(X_1 \leq x_1) P(X_2 \leq x_2) \dots P(X_n \leq x_n)$$

for all $x_1, x_2 \dots x_n \in (-\infty, \infty)$

(then conversely true)

If X_i and X_j are independent for all i and j

Throw a fair dice twice.

X_1 = Sum of the outcomes is 7

X_2 = The outcome of first throw

X_3 = " " " second "

$\Omega = \{(i,j) : 1 \le i,j \le 6\}$

$$\overbrace{P(X_2 = i, X_3 = k)}^{1/36} = \underbrace{P(X_2 = i)}_{1/6} . \underbrace{P(X_3 = k)}_{1/6}$$

X_2 and X_3 are independent

$$P(X_1 = j, X_2 = k) = \begin{cases} 0 & \text{otherwise} \\ \frac{1}{36} & X_3 = j-k \end{cases}$$

$$P(X_1 = 7, X_2 = 3, X_3 = 4) = \frac{1}{36}$$

$$\ne \underbrace{P(X_1 = 7)}_{\frac{1}{6}} . \underbrace{P(X_2 = 3) . P(X_3 = 4)}_{\frac{1}{36}}$$

Defn $X_1, X_2, \ldots$ is said to be independent sequence of r.v's if $P(X_1 \le x_1, \ldots X_n) = \prod_{i=1}^{n} P(X_i \le x_i)$ for all $-\infty \le x_1, x_2, \ldots x_n \le \infty$ and for all $n \ge 1$

Defn $X_1, X_2, \ldots$ is said to be an independent and identically distributed sequence (iid) of r.v's if

(i) $X_1, X_2, \ldots$ is an indep seq. of r.v's

(ii) for all i,j $P(X_i \le x) = P(X_j \le x)$ for all $x \in \mathbb{R}$

$$\frac{d}{ds} P_X(s)\Big|_{s=0} = p_1 = P[X=1]$$

$$\frac{d^2}{ds^2} P_X(s)\Big|_{s=0} = 2p_2 = \lfloor\underline{2}\; P[X=2]$$

$$\boxed{\frac{d^n}{ds^n} P_X(s)\Big|_{s=0} = \lfloor\underline{n}\; P[X=n]}$$

$$a = \{a_0, a_1, a_2, \dots\} \rightarrow A(s)$$
$$b = \{b_0, b_1, b_2, \dots\} \rightarrow B(s)$$

$$c_n = a_0 b_n + a_1 b_{n-1} + \dots a_{n-1} b_1 + a_n b_0$$
$$= \sum_{i=0}^{n} a_i b_{n-i}$$

$$c = \{c_0, c_1, \dots\} \longrightarrow C(s)$$

c is convolution of a and b

$$A(s)B(s) = (a_0 + a_1 s + a_2 s^2 + \dots) \times (b_0 + b_1 s + b_2 s^2 + \dots)$$
$$= a_0 b_0 + s\overbrace{(a_1 b_0 + a_0 b_1)}^{c_1} + s^2\overbrace{(a_2 b_0 + a_1 b_1 + a_0 b_2)}^{c_2} + \dots$$

$$\Rightarrow A(s)B(s) = C(s)$$

Independent $\begin{cases} X \text{ taking values } 0, 1, 2, \dots \\ Y \text{ taking values } 0, 1, 2, \dots \end{cases}$

$\Rightarrow P[X=k, Y=l] = P[X=k] . P[Y=l]$ for all k, l

$X_1 \sim Poisson(\lambda)$
$X_2 \sim Poisson(\mu)$ } indep

$$P_{X_1+X_2}(s) = P_{X_1}(s) \cdot P_{X_2}(s)$$
$$= [illegible]$$
$$= e^{-\lambda(1-s)} \cdot e^{-\mu(1-s)} = e^{-(\lambda+\mu)(1-s)}$$
$$= P_{Poisson(\lambda+\mu)}$$

$$\therefore (X_1 + X_2) \sim Poisson(\lambda+\mu)$$

$E(X^k) = m_k \rightarrow k^{th}$ moment

$$\frac{d^k}{ds^k} P_X(s) = \sum_{i=0}^{\infty} i(i-1)\dots(i-k+1) p_i s^{i-k}$$

$$\frac{d}{ds} P_X(s)\Big|_{s=1} = \sum_{i=0}^{\infty} i p_i = E(X)$$

$$\frac{d^2}{ds^2} P_X(s)\Big|_{s=1} = \sum_{i=1}^{\infty} i(i-1) p_i = E[X(X-1)]$$

$$\vdots$$

$$\frac{d^k}{ds^k} P_X(s)\Big|_{s=1} = \underbrace{E[X(X-1)(X-2)\dots(X-k+1)]}_{k^{th} \text{ factorial moment of } X}$$

$$P_X'(1) = E[X]$$
$$P_X''(1) = E[X(X-1)] = E[X^2 - X] = E[X^2] - E[X]$$
$$\Rightarrow E[X^2] = P_X''(1) + P_X'(1)$$

If X and Y are 2 rv's $P(X=k \mid Y=l)$
$$= \frac{P(X=k, Y=l)}{P(Y=l)}$$

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

__/__/____

$$Var[X] = E[X^2] - (E[X])^2$$
$$= P_X''(1) + P_X'(1) - (P_X'(1))^2$$

~~drunk~~

A person takes a step in unit time

$$X_k = \begin{cases} 0 & \text{w.p. } \frac{1}{2} \\ 1 & \text{w.p. } \frac{1}{2} \end{cases}$$

X_k : Action at $t = k$

$0 \to$ Stays, $1 \to$ moves forward

at time $= N$, he collapses

N is indep of steps

$$S_N = \sum_{i=1}^{N} X_i$$

how far he has moved

Theorem: Let $X_1, X_2, \ldots$ be iid rv's and N a rv independent of $X_1, X_2, \ldots$, Let $S_N = \sum_{i=1}^{N} X_i$

then $P_{S_N}(s) = P_N(P_{X_1}(s))$

<u>Proof</u>: $P_{S_N}(s) = \sum_{i=0}^{\infty} P(S_N = i)\, s^i$

$$= \sum_{i=0}^{\infty} s^i \sum_{j=0}^{\infty} P(S_N = i \mid N=j)\, P(N=j)$$

$$\sum_{j=0}^{\infty} P(S_N=i, N=j) = P(S_N=i)$$

$$= \sum_{i=0}^{\infty} s^i \sum_{j=0}^{\infty} P(S_N = i, N=j)$$

$$= \sum_{i=0}^{\infty} s^i \sum_{j=0}^{\infty} P\left(\sum_{l=0}^{\infty} X_l = i \mid N=j\right)$$

$$= \sum_{j=0}^{\infty} \sum_{i=0}^{\infty} s^i\, P(N=j)\, P\left(\sum_{l=0}^{\infty} X_l = i\right)$$

$$= \sum_{j=0}^{\infty} P(N=j) \sum_{i=0}^{\infty} s^i \, P\left(\sum_{l=0}^{j} X_l = i\right)$$

$$= \sum_{j=0}^{\infty} P(N=j) \, \underbrace{(P_{X_1}(s))^j}_{\to t^j}$$

$$= \sum_{j=0}^{\infty} P(N=j) \, t^j = P_N(t)$$

$$= P_N(P_X(s))$$

Example: $X_1, X_2, \ldots$ i.i.d. $X_1 = \begin{cases} 0 & \text{wp } \frac{1}{2} \\ 1 & \text{wp } \frac{1}{2} \end{cases}$

$$P_{X_1}(s) = \frac{1}{2} + \frac{1}{2}s$$

$N \sim$ geometric$\left(\frac{2}{3}\right)$

$$P_N(s) = \frac{2}{3-s}$$

$$P_{S_N}(s) = P_N(P_{X_1}(s))$$

$$= P_N\left(\frac{1+s}{2}\right) = \frac{2}{3-\left(\frac{1+s}{2}\right)} = \frac{4}{5-s}$$

__/__/__

pgf : $P_X(s) = E(s^X)$

mgf : $M_X(s) = E(e^{sX}) = \sum_{i=0}^{\infty} e^{si} P[X=i]$

$$\frac{d}{ds} M_X(s) \Big|_{s=0} = \sum_{i=0}^{\infty} i P[X=i] = E[X]$$

$$\frac{d^2}{ds^2} M_X(s) \Big|_{s=0} = \sum_{i=0}^{\infty} i^2 P[X=i] = E[X^2]$$

$$\vdots$$

$$\frac{d^k}{ds^k} M_X(s) \Big|_{s=0} = E[X^k]$$

BRANCHING PROCESS

↳ galton- watson process

\# of children of an individual is 0 w.p. $\frac{1}{2}$
1 w.p $\frac{1}{4}$
2 w.p $\frac{1}{4}$

gen 0 : 1 individual